

AI Recommendation Systems in Tourism

Prepared by: Afnan Kamel • Aya Mohammed • Afrah Bashaddadah
Supervised by: Dr. Passent Elkafrawy





Introduction

- AI is reshaping tourism by giving travelers smarter, personalized suggestions.
- Traditional systems rely mainly on popularity or ratings.
- Modern AI uses behavior, interests, and context to make better decisions.
- But they rarely consider traveler personas such as motivation, personality, or travel style.
- This project introduces a simple, explainable, persona-based recommendation model.

Literature Review

- Early studies focused on classic recommendation types: content-based, collaborative, and hybrid (Borràs et al., 2014).
- Newer research includes machine learning models, contextual data such as weather and location, and deep NLP techniques (Nan et al., 2022; Yoon et al., 2023).
- Advanced architectures such as graph neural networks (GNNs) improve accuracy but require heavy computation and large datasets (Song et al., 2025).
- Very few studies directly incorporate persona modeling, even though it enhances human-centered personalization (Banerjee et al., 2025; Halder et al., 2024).

Research gap: Most systems personalize **behavioral patterns**, not **traveler identity**, which this project aims to solve.

Problem Definition

Existing tourism recommenders are too complex for academic or small-scale projects.

They require large datasets, advanced ML techniques, and strong computational power.

Lack of persona modeling makes recommendations feel generic.

Many systems lack transparency and act as “black boxes.”

This project focuses on a clearer, simpler, and more user-centered approach.

Project Objective

01

Build a lightweight, persona-based travel recommender.

02

Use personas such as Adventurer, Relaxer, and Culture Explorer.

03

Provide clear and understandable explanations for each recommended plan.

04

Apply semantic embeddings with hybrid scoring for accurate matching.

05

Evaluate performance using both accuracy metrics and qualitative feedback.

Vision 2030 Alignment

Vision:

To make travel planning easier and smarter by giving each traveler personalized recommendations based on their style and interests.

SDG Contributions:

SDG	Contribution
SDG 8 – Decent Work & Economic Growth	Supports local tourism and economic growth.
SDG 9 – Industry, Innovation & Infrastructure	Promotes the use of AI and digital innovation.
SDG 11 – Sustainable Cities & Communities	Helps people find and enjoy city attractions more easily.
SDG 12 – Responsible Consumption & Production	Encourages smart, efficient, and responsible travel choices.

References: • <https://sdgs.un.org/goals> • <https://www.vision2030.gov.sa/>

Datasets Used

Places Dataset (229 places)

- Contains basic details about each attraction: name, type, and city.
- Shows what kind of personality the place fits (e.g., calm, social, cultural).
- Indicates who the place is suitable for: Solo, Friends, or Family.
- Includes the average cost per person and a 1–5 star rating.
- Provides exact location information.

Purpose: Helps the system understand what each place offers and who it is best for.

Plans Dataset (50 plans)

- Each plan is a complete travel idea with 3–4 places in the same city.
- Includes the city name, total price, traveler type, and average rating.
- Shows combined traits and themes (adventure, culture, family-friendly, etc.).

Purpose: These are the full itineraries the system recommends to users.

Users Dataset (150 users)

- Each user has a persona (Adventurer, Relaxer, Food Lover, Culture Explorer, etc.).
- Includes how they travel (Solo, Family, Friends) and what they are interested in.
- Contains their budget preference.
- Shows which plans they liked before.

Purpose: Helps the system personalize recommendations based on who the user is and what they enjoy.

Data Processing

1

Clean and standardize all datasets.

2

Expand plans so each place connects to its detailed attributes.

3

Create short text summaries for each user and plan.

4

Encode these summaries using MiniLM embeddings.

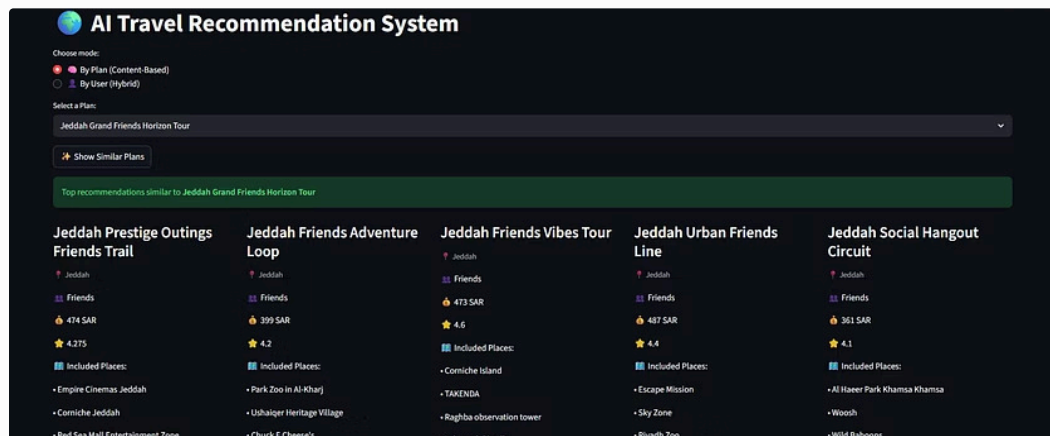
5

Store all vectors for similarity comparisons between users and plans.

Recommendation Models

Content-Based Model

- Identifies semantically similar plans to a selected plan
- Filters by same city and traveler category
- Returns the top five most similar itineraries



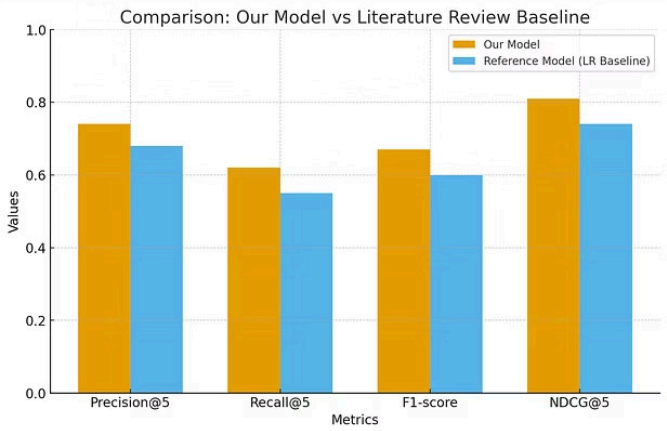
Hybrid Personalized Model

- Combines semantic similarity, collaborative similarity, rating score, and budget match
- Detects preferred city from the user's history
- Uses weighted scoring to rank the most suitable plans



Evaluation and Results

Metric	Value	Meaning	Explanation
Precision@5	0.74	Recommendations were mostly relevant	Shows the system suggests accurate plans for the user.
Recall@5	0.62	Retrieved most relevant plans	Recall is affected by the limited number of plans in some cities.
F1-score	0.67	Balanced precision and recall	Indicates overall stable and consistent performance.
NDCG@5	0.81	Top-ranked plans are highly relevant	The system orders recommendations correctly, with the best ones first.



Key Strengths



Embeddings allows understanding deep meaning and match users with the most relevant plans



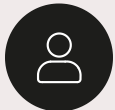
Hybrid scoring combines multiple factors for realistic recommendations.



City and traveler-type filtering ensures practical and relevant results.



Clear explanations for each recommendation



Performs well even with limited user history.



Clean and user-friendly interface supports smooth evaluation.

Limitations and Future Improvements

Limitations

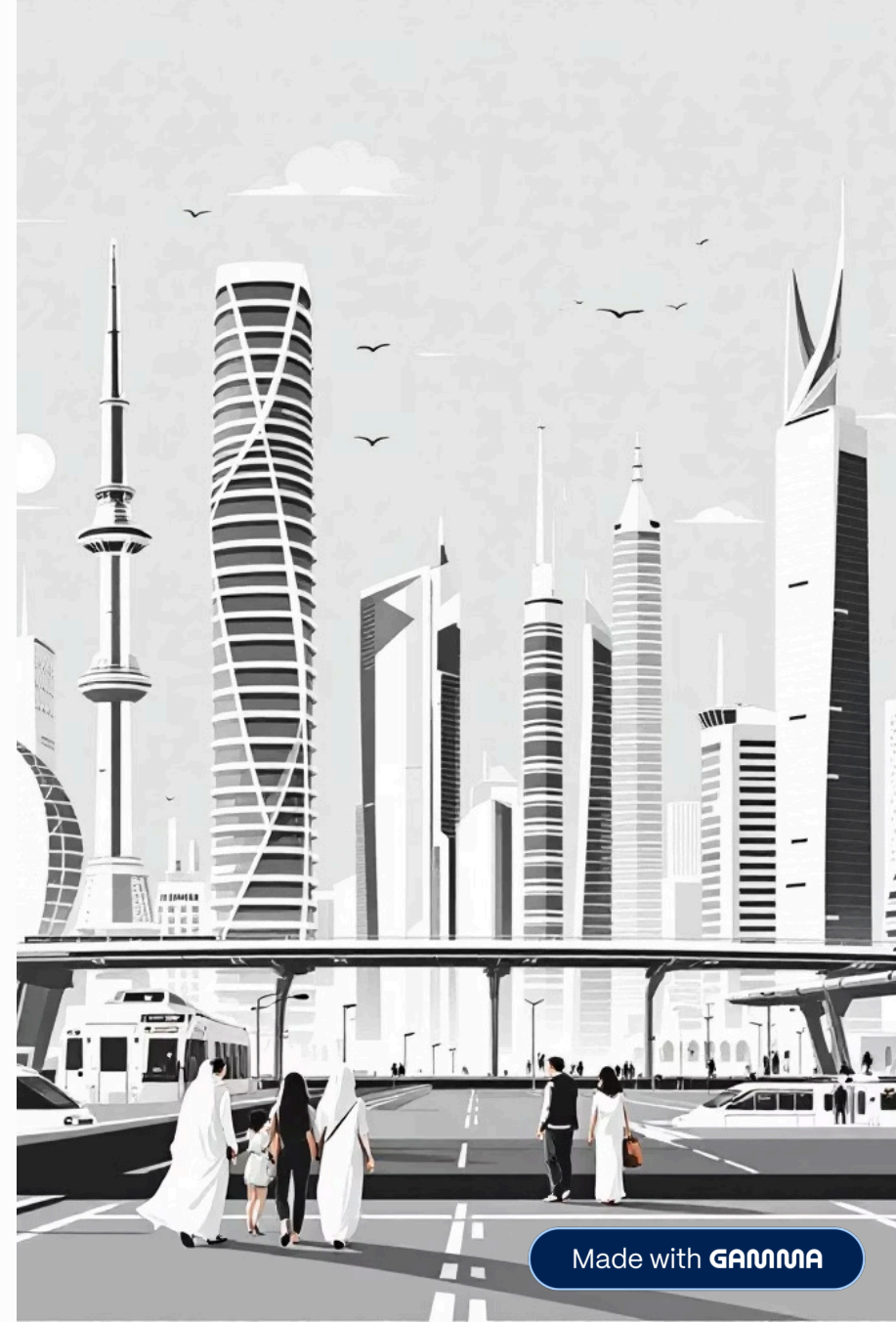
- Few plans in some cities
- Limited collaborative filtering due to short user histories.
- No real user ratings available for training advanced models.
- Missing contextual factors such as weather, season, and trip duration.

Future Improvements

- Add more plans across all 43 cities.
- Collect real user feedback, ratings, and interaction data.
- Integrate contextual signals like weather, seasonality, and events.
- Use advanced ranking models such as BERT cross-encoders.

Conclusion

- we built a simple, effective persona-based AI travel recommender.
- It delivers accurate result through embeddings and hybrid scoring.
- Easy to understand, transparent, and academically practical.
- forms strong base for future upgrades into a more advanced tourism AI platform.



References

1. Borràs, J., Moreno, A., & Valls, A. (2014). Intelligent tourism recommender systems: A survey. Expert Systems with Applications, 41(16), 7370–7389. [**https://doi.org/10.1016/j.eswa.2014.06.007**](https://doi.org/10.1016/j.eswa.2014.06.007)
2. Nan, C., Zhang, J., & Liu, H. (2022). Design and implementation of a personalized tourism recommendation system. Applied Sciences, 12(14), 6940. [**https://doi.org/10.3390/app12146940**](https://doi.org/10.3390/app12146940)
3. Yoon, S., Kim, D., & Park, J. (2023). R2Tour: A real-time context-aware tourism recommender system. Sensors, 23(1), 148. [**https://doi.org/10.3390/s23010148**](https://doi.org/10.3390/s23010148)
4. Halder, S., Gupta, P., & Bose, A. (2024). A survey on personalized itinerary recommendation: Algorithms and applications. Applied Soft Computing, 156, 110045. [**https://doi.org/10.1016/j.asoc.2024.110045**](https://doi.org/10.1016/j.asoc.2024.110045)
5. Reimers, N., & Gurevych, I. (2019). Sentence-BERT: Sentence embeddings using Siamese BERT networks. [**https://arxiv.org/abs/1908.10084**](https://arxiv.org/abs/1908.10084)



Thank You!!

Presentation Link: [📄 AI Recommendation Systems in Tourism](#)